Methods of Quantum Field Theory for T=0

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Contents

1 The Green's Function 1

1. The Green's Function

Definition 1.1 (Free-particle Green's Function)**.** The one-particle Green's function is an important quantity characterising the microscopic properties of a system defined by:

$$G_{\alpha\beta}(x,x') = -i\langle T(\tilde{\Psi}_\alpha(x)\tilde{\Psi}^\dagger_\beta(x'))\rangle \tag{1}$$

Note that x and x' denotes a set of four variables (of the coordinates $\mathbf{r}$ and t), while α and β represent spin indices.

The average value of the one-particle operator (by 6.3) is:

$$\overline{F^{(1)}} = \pm i \int \left[\lim_{\substack{\mathbf{r}'\to\mathbf{r} \\ t'\to t+0}} f^{(1)}_{\alpha\beta}(x) G_{\beta\alpha}(x,x') \right] d\mathbf{r},$$

where again the $(+)$ indicates Bose statistic; the $(-)$ indicates Fermi statistic. Note the particle (number) density; the particle current density are equal to (respectively):

$$n(x) = \pm i \lim_{\substack{\mathbf{r}'\to\mathbf{r} \\ t'\to t+0}} G_{\alpha\alpha}(x,x')$$

$$\mathbf{j}(x) = \pm \frac{1}{m} \lim_{\substack{\mathbf{r}'\to\mathbf{r} \\ t'\to t+0}} (\nabla_{\mathbf{r}} - \nabla_{\mathbf{r}'}) G_{\alpha\alpha}(x,x')$$

Another important fact about the Green's function is that it can be calculated using something called the diagram technique. We will look at all this later, but for now, we will explore some general properties of the Green's function.

In the absence of ferromagnetism and of an external magnetic field, $G_{\alpha\beta}$ takes the form:

$$G_{\alpha\beta} = G\delta_{\alpha\beta}$$

Our only focus (as of now) will be fermions. In the absence of external fields, the Green's functions only depend on coordinate differences $\mathbf{r} - \mathbf{r}'$ and time $t - t'$. Say we represent the Green's function as a Fourier integral:

$$G(x - x') = \int G(\mathbf{p},\omega) e^{i[\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}')-\omega(t-t')]} \frac{d^4p}{(2\pi)^4}, \tag{2}$$

$$d^4\mathbf{p} = d\mathbf{p}d\omega$$

To find $G(\mathbf{p},\omega)$ in a system of fermions, recall the operator:

$$\begin{cases} \tilde{\Psi}(\mathbf{r},t) = \frac{1}{\sqrt{V}}\sum_{\mathbf{p}} a_{\mathbf{p}} e^{i[\mathbf{p}\cdot\mathbf{r}-\varepsilon_0(\mathbf{p})t]} \\ \Psi(\mathbf{r},t) = \frac{1}{\sqrt{V}}\sum_{\mathbf{p}} a_{\mathbf{p}} e^{i(\mathbf{p}\cdot\mathbf{r})} \end{cases}$$

Substitution into (1), and noting that all levels $|\mathbf{p}| < p_0$ are occupied, while $|\mathbf{p}| > p_0$ is empty, we evaluate the case of $t > 0$.

Let $x' = 0$, so $G^{(0)}(x,x') = G^{(0)}(x,0) = G^{(0)}(x)$. Thus,

$$\begin{aligned} G^{(0)}(x) &= -i\langle\Phi_0|\Psi(\mathbf{r},t)\Psi^\dagger(\mathbf{r},t)|\Phi_0\rangle \\ &= -i\frac{1}{V}\sum_{\mathbf{p},\mathbf{p}'} e^{i[\mathbf{p}\cdot\mathbf{r}-\varepsilon_0(\mathbf{p})t]}\langle\Phi_0|a_{\mathbf{p}}a^\dagger_{\mathbf{p}'}|\Phi_0\rangle \\ &= -i\frac{1}{V}\sum_{\mathbf{p},\mathbf{p}'} e^{i[\mathbf{p}\cdot\mathbf{r}-\varepsilon_0(\mathbf{p})t]}\delta_{\mathbf{p}\mathbf{p}'}\langle\Phi_0|a_{\mathbf{p}}a^\dagger_{\mathbf{p}'}|\Phi_0\rangle \end{aligned}$$

By the commutation relation: $\{a_{\mathbf{p}}, a^\dagger_{\mathbf{p}}\} = 1$,

$$a_{\mathbf{p}}a^\dagger_{\mathbf{p}} = 1 - a^\dagger_{\mathbf{p}}a_{\mathbf{p}} = 1 - n_{\mathbf{p}}$$

$\therefore$ @ $t > 0$,

$$G^{(0)}(x) = \frac{-i}{V}\sum_{\mathbf{p}} e^{i[\mathbf{p}\cdot\mathbf{r}-\varepsilon_0(\mathbf{p})t]}(1-n_{\mathbf{p}})$$

For the case of $t < 0$,

$$\begin{aligned} G^{(0)}(x) &= +i\frac{1}{V}\sum_{\mathbf{p},\mathbf{p}'} e^{i[\mathbf{p}\cdot\mathbf{r}-\varepsilon_0(\mathbf{p})t]}\langle\Phi_0|a_{\mathbf{p}}a^\dagger_{\mathbf{p}'}|\Phi_0\rangle \\ &= +i\frac{1}{V}\sum_{\mathbf{p},\mathbf{p}'} e^{i[\mathbf{p}\cdot\mathbf{r}-\varepsilon_0(\mathbf{p})t]}n_{\mathbf{p}} \\ &= \frac{-i}{V}\sum_{\mathbf{p}} e^{i[\mathbf{p}\cdot\mathbf{r}-\varepsilon_0(\mathbf{p})t]}(-n_{\mathbf{p}}) \end{aligned}$$

To conclude,

$$G^{(0)}(x) = \frac{-i}{V}\sum_{\mathbf{p}} e^{i[\mathbf{p}\cdot\mathbf{r}-\varepsilon_0(\mathbf{p})t]}\begin{cases} 1-n_{\mathbf{p}}, & t>0 \\ -n_{\mathbf{p}}, & t<0 \end{cases} \tag{3}$$

where:

$$n_{\mathbf{p}} = a^\dagger_{\mathbf{p}}a_{\mathbf{p}} = \begin{cases} 1, & |\mathbf{p}| < p_0 \\ 0, & |\mathbf{p}| > p_0 \end{cases}$$

In the momentum representation, we can take the Fourier transform of respect to $\mathbf{r}$ and t.

Let:

$$G^{(0)}(\mathbf{p},\omega) = \int_{-\infty}^{+\infty} dt \int d^3r\, G^{(0)}(\mathbf{r},t)e^{-i(\mathbf{p}\cdot\mathbf{r}-\omega t)}$$

Substituting (3) yields:

$$G^{(0)}(\mathbf{p},\omega) = \int_{-\infty}^{+\infty} dt \int d^3r \left[\frac{-i}{V} \sum_{\mathbf{p}'} e^{i[\mathbf{p}'\cdot\mathbf{r} - \varepsilon_0(\mathbf{p}')t]} \begin{cases} 1 - n_{\mathbf{p}'}, & t > 0 \\ -n_{\mathbf{p}'}, & t < 0 \end{cases} \right] e^{-i(\mathbf{p}\cdot\mathbf{r} - \omega t)}$$

Grouping like terms (that are spatially dependent),

$$G^{(0)}(\mathbf{p},\omega) = \frac{-i}{V} \int_{-\infty}^{+\infty} dt \sum_{\mathbf{p}'} e^{-i\varepsilon_0(\mathbf{p}')t} \begin{cases} 1 - n_{\mathbf{p}'}, & t > 0 \\ -n_{\mathbf{p}'}, & t < 0 \end{cases} e^{i\omega t} \int d^3r \, e^{i(\mathbf{p}' - \mathbf{p})\cdot\mathbf{r}}$$

To evaluate the spatial integral, since we know plane waves are orthogonal in an enclosed volume,

$$\int d^3r \, e^{i(\mathbf{p}' - \mathbf{p})\cdot\mathbf{r}} = V\delta_{\mathbf{p}\mathbf{p}'}$$

Substitution of this expression into the main integrand,

$$\begin{aligned} G^{(0)}(\mathbf{p},\omega) &= \frac{-i}{V} \int_{-\infty}^{+\infty} dt \sum_{\mathbf{p}'} e^{-i\varepsilon_0(\mathbf{p}')t} \begin{cases} 1 - n_{\mathbf{p}'}, & t > 0 \\ -n_{\mathbf{p}'}, & t < 0 \end{cases} e^{i\omega t} (V\delta_{\mathbf{p}\mathbf{p}'}) \\ &= -i \int_{-\infty}^{+\infty} dt \sum_{\mathbf{p}'} e^{-i\varepsilon_0(\mathbf{p}')t} \begin{cases} 1 - n_{\mathbf{p}'}, & t > 0 \\ -n_{\mathbf{p}'}, & t < 0 \end{cases} e^{i\omega t} \delta_{\mathbf{p}\mathbf{p}'} \\ &= -i \left[\int_0^{+\infty} dt \, e^{i[\omega - \varepsilon_0(\mathbf{p})]t} (1 - n_{\mathbf{p}}) + \int_{-\infty}^{0} dt \, e^{i[\omega - \varepsilon_0(\mathbf{p})]t} (-n_{\mathbf{p}}) \right] \end{aligned}$$

Alternatively, we write this as:

$$G^{(0)}(\mathbf{p},\omega) = -i \left\{ \theta(|\mathbf{p}| - p_0) \int_0^{+\infty} e^{i[\omega - \varepsilon_0(\mathbf{p})]t} dt - \theta(p_0 - |\mathbf{p}|) \int_{-\infty}^{0} e^{i[\omega - \varepsilon_0(\mathbf{p})]t} dt \right\}, \quad \ldots (4) \quad (4)$$

where

$$\theta(z) = \begin{cases} 1, & z > 0 \\ 0, & z < 0 \end{cases}$$

The expression for $G^{(0)}(\mathbf{p},\omega)$ contains two integrals of the type:

$$\int_0^{\infty} e^{ist} dt$$

An integral of this type is defined as the limit:

$$\lim_{\delta \to +0} \int_0^{\infty} e^{ist - \delta t} dt = i \lim_{\delta \to +0} \frac{1}{s + i\delta} \quad (5)$$

The term $i\delta$ in the denominator bypasses the pole (point where function blows up to infinity when denominator equals zero) @ $s = 0$ in evaluating integrals of a function, namely:

$$\int \frac{F(s)\, ds}{s + i\delta} = P \int \frac{F(s)}{s} \, ds - i\pi F(0),$$

where P is the operation of taking the principal value of an integral. We can more formally express this as:

$$\frac{1}{s+i\delta} = \frac{P}{s} - i\pi\delta(s).$$

A sleight of notation is that sometimes, $\delta_+(s)$ is used to denote the quantity

$$\frac{1}{\pi}\frac{i}{s+i\delta},$$

i.e.,

$$\delta_+(s) = \delta(s) - \frac{P}{i\pi s} \tag{6}$$

By (4) and (5), we obtain:

$$G^{(0)}(\mathbf{p},\omega) = \frac{\theta(|\mathbf{p}|-p_0)}{\omega-\varepsilon_0(\mathbf{p})+i\delta} + \frac{\theta(p_0-|\mathbf{p}|)}{\omega-\varepsilon_0(\mathbf{p})-i\delta}$$

Notice that there is only a sign change in terms of δ, so:

$$G^{(0)}(\mathbf{p},\omega) = \frac{1}{\omega-\varepsilon_0(\mathbf{p})+i\delta\,\mathrm{sgn}(|\mathbf{p}|-p_0)} \tag{7}$$

Now, lets consider the simplest case of phonons. The displacement of points are denoted by $\mathbf{q}(\mathbf{r},t)$. Then, momentum is denoted as: $\rho\dot{\mathbf{q}}(\mathbf{r},t)$, ρ being the density. According to quantum mechanics, quantities $\mathbf{q}$ and $\dot{\mathbf{q}}$ are operators of the commutation rules:

$$\rho[\dot{q}_i(\mathbf{r},t), q_j(\mathbf{r}',t)] = -i\delta(\mathbf{r}-\mathbf{r}')\delta_{ij} \tag{8}$$

Expanding $\mathbf{q}$ in plane waves. To note the value of the wave vector $\mathbf{k}$ uniquely determines the frequency, which we denote by $\omega_0(\mathbf{k})$. Thus,

$$\mathbf{q}(\mathbf{r},t) = \frac{1}{\sqrt{V}}\sum_{\mathbf{k}}\frac{\mathbf{k}}{|\mathbf{k}|}\left\{q_{\mathbf{k}}e^{i[\mathbf{k}\cdot\mathbf{r}-\omega_0(\mathbf{k})t]} + q_{\mathbf{k}}^{\dagger}e^{-i[\mathbf{k}\cdot\mathbf{r}-\omega_0(\mathbf{k})t]}\right\} \tag{9}$$

We now introduce an operator $b_{\mathbf{k}}$ that is related to $q_{\mathbf{k}}$ by the following formula:

$$q_{\mathbf{k}} = \frac{b_{\mathbf{k}}}{\sqrt{2\rho\omega_0(\mathbf{k})}} \tag{10}$$

The kinetic energy operator is:

$$K = \frac{\rho}{2}\int\left[\dot{\mathbf{q}}(\mathbf{r},t)\right]^2 d\mathbf{r} \tag{11}$$

Applying virial theorem, we obtain the formula:

$$H = 2K = \sum_{\mathbf{k}}\omega_0(\mathbf{k})(n_{\mathbf{k}}+1/2) \tag{12}$$

where $n_{\mathbf{k}} = b_{\mathbf{k}}^{\dagger}b_{\mathbf{k}}$.

The operators $\mathbf{q}$ can be more conveniently defined to anticipate the interaction of phonons and electrons in a metal:

$$\varphi(x) = \frac{i}{\sqrt{V}}\sum_{\mathbf{k}}\sqrt{\frac{\omega_0(\mathbf{k})}{2}}\left\{b_{\mathbf{k}}e^{i[\mathbf{k}\cdot\mathbf{r}-\omega_0(\mathbf{k})t]} - b_{\mathbf{k}}^{\dagger}e^{-i[\mathbf{k}\cdot\mathbf{r}-\omega_0(\mathbf{k})t]}\right\} \tag{13}$$

This formula is specific to longitudinal phonons in the Debye model. The Green's function of phonons is usually denoted by the letter D.

$$D(x, x') = -i\langle T(\varphi(x)\varphi(x'))\rangle \tag{14}$$

Substitution of (13) into (14) yields:

$$D^{(0)}(x) = \frac{-i}{V}\sum_{\mathbf{k}} \frac{\omega_0(\mathbf{k})}{2} \begin{cases} e^{i[\mathbf{k}\cdot\mathbf{r}-\omega_0(\mathbf{k})t]}, & t > 0 \\ e^{-i[\mathbf{k}\cdot\mathbf{r}-\omega_0(\mathbf{k})t]}, & t < 0 \end{cases} \tag{15}$$

The Fourier components of (15):

$$\begin{aligned} D^{(0)}(\mathbf{k}, \omega) &= \frac{\omega_0(\mathbf{k})}{2}\left[\frac{1}{\omega - \omega_0(\mathbf{k}) + i\delta} - \frac{1}{\omega + \omega_0(\mathbf{k}) - i\delta}\right] \\ &= \frac{\omega_0^2(\mathbf{k})}{\omega^2 - \omega_0^2(\mathbf{k}) + i\delta} \end{aligned} \tag{16}$$

References

[1] A. A. Abrikosov, L. P. Gorkov, and I. E. Dzyaloshinski, *Methods of Quantum Field Theory in Statistical Physics*. New York, NY: Dover Publications, 1975.